Applied Econometrics

Model Selection & Diagnostic Checklist

A Practical Reference for Researchers

When to use each model · Key assumptions · Diagnostic tests
Decision rules · Common pitfalls · Step-by-step checklists

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Python for Econometrics Course

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1. Quick Model-Selection Decision Tree

Step 1 — What is your data structure?

- Cross-sectional (one observation per unit) → Sections 2–7
- Time series (one unit over time) → Sections 8–10
- Panel (multiple units over time) → Section 11
- Binary / limited outcome → Section 13

Step 2 — Is the outcome continuous and unbounded?

Yes → OLS family. No (binary/count/censored) → Section 13.

Step 3 — Is your key regressor endogenous?

Suspect endogeneity (OVB, measurement error, simultaneity)? → Section 12 (IV/2SLS) or Section 14 (Treatment Effects).

Step 4 — Run OLS, then check diagnostics in order:

Multicollinearity (Sec. 3) → Heteroscedasticity (Sec. 4) → Specification (Sec. 5) → Serial correlation if time series (Sec. 9) → Stationarity if time series (Sec. 8, 10).

Setting	First choice	If endogeneity	If binary Y	If non-stationary
Cross-section	OLS	IV / 2SLS	Logit / Probit	—
Time series	OLS + HAC	IV	—	ARIMA / ECM
Panel	FE / RE	FE + IV	—	—
Program eval.	DiD / RDD	IV-LATE	—	—

2. OLS — Simple & Multiple Regression

When to Use

- Continuous, unbounded dependent variable
- Exploring linear relationships between variables
- Baseline before testing for violations
- Gauss–Markov conditions plausibly satisfied

Watch Out For

- Omitted variable bias: Bias $= \beta_2 \cdot \delta_1$
- Functional form misspecification (try logs, quadratics)
- Influential outliers (check Cook's distance)
- Extrapolation beyond data support
- Interpreting R^2 as causal fit

Key Formula / Estimator

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

$$\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$$

BLUE when MLR.1–MLR.5 hold (Gauss–Markov Theorem).

Diagnostic Tests & Decision Rules

Test	Null hypothesis	Decision
t-test	$H_0 : \beta_j = 0$	$\lvert t\rvert > t_{\alpha/2,n-k-1} \Rightarrow$ **reject H_0**
F-test (joint)	H_0 : all slope $\beta = 0$	$F > F_{\alpha,k,n-k-1} \Rightarrow$ **reject H_0**
RESET test	H_0 : functional form OK	p-value ¡ 0.05 $\Rightarrow$ misspecification
Partial F-test	H_0 : group of βs $= 0$	$F > F_{\alpha,q,n-k-1} \Rightarrow$ **reject H_0**

Researcher Checklist

- ☐ State the population model explicitly
- ☐ Check variable distributions (histograms)
- ☐ Consider log transformation for skewed vars
- ☐ Run OLS, inspect coefficients and signs
- ☐ Check R^2 and adjusted R^2
- ☐ Run RESET test for functional form
- ☐ Inspect residual plot vs. fitted values
- ☐ Run joint F-test for overall significance
- ☐ Check for influential observations
- ☐ Interpret coefficients as *ceteris paribus*

3. Multicollinearity

When to Use

- After OLS, if SEs are large / signs unexpected
- Multiple regressors that may be highly correlated
- Model includes polynomial or interaction terms
- Structural economic variables (income, wealth, consumption)

Watch Out For

- Inflated SEs → false non-significance
- Coefficients sensitive to adding/dropping vars
- Signs may flip relative to bivariate correlations
- High R^2 with few significant t-tests
- **Not a violation** of Gauss–Markov; OLS still BLUE

Key Formula / Estimator

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

where $R_j^2 = R^2$ from regressing x_j on all other regressors.
Condition number: $\kappa = \sqrt{\lambda_{\max}/\lambda_{\min}}$

Diagnostic Tests & Decision Rules

Indicator	What it measures	Threshold
VIF_j	Variance inflation for $\hat{\beta}_j$	> 10 severe, > 5 moderate
Condition number κ	Overall collinearity	> 30 severe
Pairwise r	Simple correlation	$\lvert r\rvert > 0.8$ flag
Eigenvalue ratio	Near-zero eigenvalues	$\lambda < 0.01$ flag

Researcher Checklist

- ☐ Compute pairwise correlations of regressors
- ☐ Compute VIF for each regressor
- ☐ Check condition number of $\mathbf{X}'\mathbf{X}$
- ☐ If VIF > 10: consider dropping one variable
- ☐ Consider combining correlated vars (index/PCA)
- ☐ Try Ridge if prediction is the goal
- ☐ Report VIFs in results table
- ☐ Do not drop variables with theoretical justification

4. Heteroscedasticity

When to Use

- Cross-sectional data with varying unit sizes (firms, countries)
- Income/expenditure data (variance rises with level)
- After OLS, residuals show fan/funnel pattern
- Any time you want valid inference from OLS

Watch Out For

- OLS coefficients still unbiased, SEs wrong
- t and F statistics invalid without correction
- White test has low power in small samples
- WLS requires correct variance specification
- Always use robust SEs by default in cross-section

Key Formula / Estimator

Heteroscedasticity-robust (HC1) SEs:

$$\widehat{\text{Var}}_{\text{HC1}} = \frac{n}{n-k-1}(\mathbf{X}'\mathbf{X})^{-1}\left(\sum_i \hat{u}_i^2 \mathbf{x}_i \mathbf{x}_i'\right)(\mathbf{X}'\mathbf{X})^{-1}$$

WLS: weight by $w_i = 1/\hat{\sigma}_i^2$.

Diagnostic Tests & Decision Rules

Test	H_0	Statistic	Decision
Breusch–Pagan	Homoscedastic	$nR^2 \sim \chi^2_k$	$p < 0.05 \Rightarrow$ **reject H_0**
White	Homoscedastic	$nR^2 \sim \chi^2_p$	$p < 0.05 \Rightarrow$ **reject H_0**
Goldfeld–Quandt	Homoscedastic	F-ratio of variances	$p < 0.05 \Rightarrow$ **reject H_0**

Remedy hierarchy: (1) Use HC1/HC3 robust SEs → (2) WLS if variance structure known → (3) Log-transform y if level variance rises with level.

Researcher Checklist

- ☐ Plot residuals vs. fitted values
- ☐ Plot $|\hat{u}|$ or $\hat{u}^2$ vs. each regressor
- ☐ Run Breusch–Pagan test
- ☐ Run White test (more general)
- ☐ If tests reject: use `cov_type='HC1'`
- ☐ If variance structure known: use WLS
- ☐ Compare OLS vs. robust SEs in table
- ☐ Check if log-transform eliminates problem

5. Model Specification & Data Issues

When to Use

- Always: before trusting OLS results
- When theory suggests non-linear relationships
- When proxy variables used (e.g., IQ for ability)
- When measurement error in key regressor suspected

Watch Out For

- Measurement error biases $\hat{\beta}$ toward zero
- Non-linearity looks like heteroscedasticity
- Proxy variables introduce attenuation
- Outliers can drive OLS results in small samples
- Missing variable vs. irrelevant variable trade-off

Key Formula / Estimator

Attenuation bias (measurement error):

$$\text{plim}(\hat{\beta}_1) = \beta_1 \cdot \frac{\sigma^2_{x^*}}{\sigma^2_{x^*} + \sigma^2_v}$$

RESET statistic: Add $\hat{y}^2, \hat{y}^3$ to regression; test their joint significance.

Researcher Checklist

- ☐ Run RESET test ($p < 0.05 \rightarrow$ misspecification)
- ☐ Try log, squared, interaction terms if RESET fails
- ☐ Check for outliers; run model with and without
- ☐ Document proxy variable limitations
- ☐ If meas. error: consider IV for the mismeasured var
- ☐ Davidson–MacKinnon J-test for non-nested models
- ☐ Report AIC / BIC for model comparison
- ☐ Keep all theoretically justified variables

6. Qualitative Regressors & Interaction Effects

When to Use

- Categorical variables (gender, region, industry)
- Testing if slopes differ across groups
- Policy variables (treatment/control indicators)
- Chow test for structural stability

Watch Out For

- Always omit one category (base group)
- Interaction terms amplify multicollinearity
- Centre continuous vars before interacting
- Chow test assumes homoscedasticity
- Saturated models with many dummies: overfit risk

Key Formula / Estimator

Dummy variable trap: include $G-1$ dummies for G categories.

Interaction: $y = \beta_0 + \beta_1 d + \beta_2 x + \beta_3 d \cdot x + u$
β_3 = difference in slope between groups.

Chow test: $F = \frac{(RSS_R - RSS_U)/k}{RSS_U/(n-2k)}$

Researcher Checklist

- ☐ Choose interpretable base/reference category
- ☐ Include $G-1$ dummies; verify no perfect collinearity
- ☐ Test significance of dummy coefficient
- ☐ Test interaction term if slopes may differ
- ☐ Chow test if structural break suspected
- ☐ Report marginal effect at meaningful base group
- ☐ Plot fitted values by group for visual check
- ☐ Label coefficients clearly in output

7. Endogeneity & Instrumental Variables

When to Use

- OVB suspected (ability, motivation, quality)
- Simultaneity (supply/demand, wages/education)
- Measurement error in key regressor
- Natural experiment or policy instrument available
- Durbin–Wu–Hausman test rejects exogeneity

Watch Out For

- Weak instruments ($F_{\text{1st stage}} < 10$) inflate 2SLS bias
- Instrument must be exogenous (untestable with one IV)
- 2SLS SEs larger than OLS by construction
- Overidentification: test exclusion with Sargan/Hansen
- LATE ≠ ATE: IV estimates effect for compliers only
- 2SLS inconsistent if instrument invalid

Key Formula / Estimator

OVB formula: $\text{plim}(\tilde{\beta}_1) = \beta_1 + \beta_2 \delta_1$
where $\delta_1 = \text{Cov}(x_2, x_1)/\text{Var}(x_1)$.

2SLS estimator:

$$\hat{\boldsymbol{\beta}}_{2SLS} = (\hat{\mathbf{X}}'\mathbf{X})^{-1}\hat{\mathbf{X}}'\mathbf{y}$$

First stage: $x = \pi_0 + \pi_1 z + v$, need $\pi_1 \neq 0$.

Diagnostic Tests & Decision Rules

Test	Purpose	Decision
First-stage F	Instrument strength	$F > 10$ (Stock–Yogo rule)
Durbin–Wu–Hausman	Endogeneity of regressor	$p < 0.05 \Rightarrow$ use IV
Sargan/Hansen J	Overidentification (validity)	$p < 0.05 \Rightarrow$ suspect IV
Partial R^2 (1st stage)	Instrument relevance	Higher is better

Researcher Checklist

- ☐ State economic argument for endogeneity
- ☐ Identify candidate instruments (theory-based)
- ☐ Check instrument relevance: 1st-stage $F > 10$
- ☐ Argue instrument exogeneity (no direct effect on y)
- ☐ Run Durbin–Wu–Hausman test (compare OLS vs 2SLS)
- ☐ If > 1 instrument: run Sargan/Hansen J-test
- ☐ Report both OLS and 2SLS side by side
- ☐ Interpret 2SLS as LATE if instrument is binary

8. Panel Data: Fixed Effects, Random Effects & First Differences

When to Use

- Data: multiple units observed over multiple periods
- Control for unobserved unit-level heterogeneity
- FE: if unobserved effect correlated with regressors
- RE: if unobserved effect uncorrelated (efficiency gain)
- FD: if $T = 2$ or non-stationary panel

Key Formula / Estimator

FE (within estimator): demean all variables: $\ddot{y}_{it} = \ddot{\mathbf{x}}_{it}\boldsymbol{\beta} + \ddot{u}_{it}$

RE: GLS with $\lambda = 1 - \sigma_u / \sqrt{\sigma_u^2 + T\sigma_a^2}$

FD: $\Delta y_{it} = \Delta \mathbf{x}_{it}\boldsymbol{\beta} + \Delta u_{it}$

Watch Out For

- FE removes all time-invariant variables (including gender, race)
- FE amplifies attenuation bias from meas. error
- RE inconsistent if $\text{Cov}(a_i, x_{it}) \neq 0$
- Cluster SEs by unit if serial correlation in u_{it}
- FD loses one time period; FE more efficient with $T > 2$
- Time dummies needed to control aggregate shocks

Diagnostic Tests & Decision Rules

Test	H_0	Decision
Hausman test	RE consistent ($\alpha_i \perp x_{it}$)	$p < 0.05 \Rightarrow$ use FE
F-test for FE	All $\alpha_i = 0$ (pooled OLS OK)	$p < 0.05 \Rightarrow$ use FE
Breusch–Pagan LM	$\text{Var}(\alpha_i) = 0$ (pooled OK)	$p < 0.05 \Rightarrow$ use RE
Serial corr. (Wooldridge)	No AR(1) in u_{it}	$p < 0.05 \Rightarrow$ cluster SEs

Researcher Checklist

- ☐ Identify panel structure: balanced or unbalanced?
- ☐ Run pooled OLS as baseline
- ☐ Run Breusch–Pagan LM test (pool vs RE)
- ☐ Run FE and RE; save both sets of estimates
- ☐ Run Hausman test: $p < 0.05 \rightarrow$ FE
- ☐ Add time dummies (test jointly with F-test)
- ☐ Cluster SEs by unit (panel robust)
- ☐ FE cannot identify time-invariant vars: plan accordingly

9. Stationary Time Series

When to Use

- Confirmed stationary series (ADF rejects unit root)
- Forecasting or modelling dynamic relationships
- Distributed lag models (PDL, ADL)
- Causal analysis between time series

Key Formula / Estimator

ADL(p,q) model: $y_t = \alpha + \sum_{j=1}^{p} \rho_j y_{t-j} + \sum_{s=0}^{q} \delta_s x_{t-s} + u_t$

Long-run multiplier: $\frac{\sum \delta_s}{1 - \sum \rho_j}$

HAC (Newey–West) SEs correct for serial correlation.

Watch Out For

- Trend-stationary vs. difference-stationary: different treatment
- Seasonality: add seasonal dummies or use seasonal differencing
- Spurious regression if levels used on I(1) series
- Lag selection: AIC/BIC; don't over-fit
- Structural breaks violate stationarity assumption
- Granger causality ≠ economic causality

Researcher Checklist

- ☐ Plot time series; look for trend, seasonality
- ☐ Run ADF test (with and without trend)
- ☐ If I(0): proceed with levels
- ☐ Include trend/seasonal dummies if needed
- ☐ Select lag length via AIC or BIC
- ☐ Use Newey–West (HAC) SEs for inference
- ☐ Check residuals for remaining autocorrelation (BG)
- ☐ Granger causality test if dynamic relationship of interest

10. Serial Correlation

When to Use

- Time series or panel data residuals
- OLS residuals show pattern (ACF plot, DW $\neq 2$)
- Dynamically misspecified model
- Economic data with inertia (GDP, unemployment)

Key Formula / Estimator

AR(1): $u_t = \rho u_{t-1} + e_t,\ |\rho| < 1$

DW statistic: $DW \approx 2(1-\hat{\rho})$
$DW \approx 2$: no autocorrelation

Cochrane–Orcutt: transform by $y_t - \hat{\rho} y_{t-1}$ and re-estimate.

Watch Out For

- OLS unbiased but inefficient; SEs biased downward
- DW test invalid with lagged y as regressor (use BG)
- Serial correlation often symptom of omitted dynamics
- First fix: add lagged y or lagged x before using HAC
- In panels: cluster SEs by unit and period

Diagnostic Tests & Decision Rules

Test	H_0	Decision
Durbin–Watson	$\rho = 0$ (no AR(1))	$DW < d_L \Rightarrow$ **reject H_0**; avoid if lagged y
Breusch–Godfrey	H_0: no AR(p)	$p < 0.05 \Rightarrow$ **reject H_0**
ACF/PACF plot	Visual check	Spikes outside $\pm 1.96/\sqrt{T}$
Ljung–Box Q	No autocorrelation to lag h	$p < 0.05 \Rightarrow$ **reject H_0**

Researcher Checklist

- ☐ Plot ACF and PACF of residuals
- ☐ Run Breusch–Godfrey test (BG, lag 1 and 4)
- ☐ If dynamic model: BG preferred over DW
- ☐ First remedy: add dynamic terms (lags)
- ☐ If persist: use Newey–West HAC SEs (`cov_type='HAC'`)
- ☐ Cochrane–Orcutt for GLS if AR(1) confirmed
- ☐ After remedy: re-run BG to verify
- ☐ Never use DW with lagged dependent variable

11. Non-Stationary Time Series: Unit Roots & Cointegration

When to Use

- ADF fails to reject unit root (series is I(1))
- Macro series: GDP, prices, exchange rates
- Long-run equilibrium relationship between I(1) series
- Error correction model for short-run adjustment

Key Formula / Estimator

ADF regression:
$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{j=1}^{p} \delta_j \Delta y_{t-j} + u_t$
H_0: $\gamma = 0$ (unit root).

ECM: $\Delta y_t = \alpha + \lambda \hat{e}_{t-1} + \beta \Delta x_t + u_t$
$\hat{\lambda} < 0$: error correction toward equilibrium.

Watch Out For

- Spurious regression: high R^2, low DW with I(1) series
- First-differencing removes long-run information
- Cointegration required before ECM is valid
- ADF has low power in small samples (PP test as check)
- Include intercept/trend in ADF based on data plot
- Johansen for multivariate cointegration

Diagnostic Tests & Decision Rules

Test	H_0	Decision
ADF	Unit root ($\gamma = 0$)	$p < 0.05 \Rightarrow$ stationary
Phillips–Perron	Unit root (non-parametric)	$p < 0.05 \Rightarrow$ stationary
Engle–Granger	No cointegration (resid I(1))	$p < 0.05 \Rightarrow$ cointegrated
Johansen trace	$\leq r$ cointegrating vecs	Reject if stat $>$ critical value

Researcher Checklist

- ☐ Plot each series; visual check for trend/drift
- ☐ Run ADF (with constant and with trend)
- ☐ Also run PP test as robustness check
- ☐ If I(1): check cointegration (Engle–Granger or Johansen)
- ☐ If cointegrated: estimate ECM; verify $\hat{\lambda} < 0$
- ☐ If not cointegrated: work in first differences
- ☐ Report DW in levels regression (near 0 = spurious)
- ☐ Never regress I(1) on I(1) without checking cointegration

12. Difference-in-Differences & Pooled Cross-Sections

When to Use

- Natural experiment: policy affects one group, not another
- Pre- and post-treatment observations available
- Treatment assignment not random but parallel trends plausible
- Pooled cross-sections: repeated surveys (not panel)

Key Formula / Estimator

$$\hat{\delta}_{\text{DiD}} = (\bar{y}_{T,\text{after}} - \bar{y}_{T,\text{before}}) - (\bar{y}_{C,\text{after}} - \bar{y}_{C,\text{before}})$$

Regression form:
$y = \beta_0 + \beta_1 D_{\text{treat}} + \beta_2 D_{\text{post}} + \delta D_{\text{treat}} \cdot D_{\text{post}} + u$
δ = ATT under parallel trends.

Watch Out For

- Parallel trends is **untestable** post-treatment (test pre-trends)
- Spillover effects violate SUTVA
- Staggered adoption: use Callaway–Sant'Anna, not simple DiD
- Always cluster SEs at group (treated/control) level
- Anticipation effects can bias DiD
- Event study plots: check pre-trend coefficients near zero

Researcher Checklist

- ☐ Plot parallel trends in pre-period (visual + event study)
- ☐ Define treatment/control groups clearly
- ☐ Verify no anticipation effects before treatment
- ☐ Run full DiD regression with interaction term
- ☐ Cluster SEs by group or state
- ☐ Add controls to improve precision (not for identification)
- ☐ Check heterogeneous treatment effects
- ☐ If staggered: use modern DiD estimator

13. Treatment Effects & Causal Inference

When to Use

- Estimating causal effect of program/policy/treatment
- RCT: randomised control trial available
- DiD: quasi-experiment with parallel trends
- RDD: assignment determined by threshold rule
- PSM: observational data, selection on observables

Key Formula / Estimator

Potential outcomes: $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$
$\text{ATE} = E[Y(1) - Y(0)]$
$\text{ATT} = E[Y(1) - Y(0) \mid D = 1]$
$\text{LATE} = \frac{\text{Reduced form}}{\text{First stage}}$ (IV for compliers)
Naive bias: $\hat{\tau}_{\text{naive}} = \text{ATT} + \underbrace{E[Y(0) \mid D{=}1] - E[Y(0) \mid D{=}0]}_{\text{selection bias}}$

Watch Out For

- RCT: OLS on indicator gives ATE unbiasedly
- PSM: only controls for *observed* confounders
- RDD: local estimate near threshold; check manipulation
- LATE $\neq$ ATE unless instrument affects all units (strong)
- Always report which estimand (ATE/ATT/LATE) you target
- Large bandwidth in RDD: more data, more bias
- Common support must hold for PSM (check overlap)

Diagnostic Tests & Decision Rules

Test / Check	Purpose	Decision
Balance table	Covariate balance post-randomisation	All $p > 0.05$
McCrary density test	No manipulation at RDD cutoff	$p > 0.05$ OK
Propensity overlap plot	Common support for PSM	Check visually
Placebo cutoff (RDD)	No effect away from threshold	$p > 0.05$ OK
Pre-trend test (DiD)	Parallel pre-trends	Coeff near 0

Researcher Checklist

- ☐ State causal question and target estimand (ATE/ATT/LATE)
- ☐ Choose identification strategy based on data structure
- ☐ RCT: verify randomisation (balance table)
- ☐ DiD: test parallel pre-trends (event study)
- ☐ RDD: McCrary test; vary bandwidth; placebo cutoff
- ☐ PSM: estimate propensity score; check overlap
- ☐ IV/LATE: 1st-stage $F > 10$; instrument exogeneity
- ☐ Report estimand, sample, and assumptions explicitly

14. Limited Dependent Variables: Logit, Probit & Tobit

When to Use

- **Logit/Probit:** binary outcome $y \in \{0, 1\}$
- **Tobit:** censored/truncated outcome ($y \geq 0$, many zeros)
- **Ordered Probit:** ordinal outcome (Likert scales)
- **Multinomial Logit:** unordered multiple categories

Key Formula / Estimator

Logit: $P(y = 1|\mathbf{x}) = \Lambda(\mathbf{x}\boldsymbol{\beta}) = \frac{e^{\mathbf{x}\boldsymbol{\beta}}}{1+e^{\mathbf{x}\boldsymbol{\beta}}}$

Marginal effect (AME): $\frac{\partial P}{\partial x_j} = \hat{\beta}_j \cdot \bar{\lambda}$

where $\bar{\lambda}$ = avg. density evaluated at $\hat{\mathbf{x}}\hat{\boldsymbol{\beta}}$.

Probit: Φ replaces Λ; coefficients $\approx \beta_{\text{logit}}/1.6$.

Watch Out For

- LPM (OLS on binary y): predicted probs outside [0,1]
- Logit/Probit coefficients: direction only, not magnitude
- Always compute and report marginal effects (AME/MEM)
- Pseudo-R^2 (McFadden) not comparable to OLS R^2
- Tobit assumes normality; check with residual plots
- Perfect separation (predictor fully determines y): MLE fails

Diagnostic Tests & Decision Rules

Test	Purpose	Decision
LR test	Joint significance of regressors	$p < 0.05 \Rightarrow$ **reject H_0**
Hosmer–Lemeshow	Goodness of fit (calibration)	$p > 0.05$ good fit
Wald test	Individual coefficient $= 0$	$p < 0.05$ significant
AUC / ROC	Discrimination ability	> 0.7 acceptable
Vuong test	Logit vs. Probit model choice	Significant $\rightarrow$ prefer one

Researcher Checklist

- ☐ Confirm y is binary/limited before logit/probit
- ☐ Estimate logit and probit; compare signs/significance
- ☐ Always compute Average Marginal Effects (AME)
- ☐ Report pseudo-R^2 and classification accuracy
- ☐ Check for perfect or quasi-separation
- ☐ Hosmer–Lemeshow test for calibration
- ☐ Plot ROC curve and report AUC
- ☐ Do not report raw logit/probit coefficients as effects

15. Advanced Time Series: ARIMA, VAR & Granger Causality

When to Use

- Univariate forecasting with autocorrelated errors: ARIMA
- System of mutually dependent time series: VAR
- Testing predictive causality between series: Granger
- Impulse response analysis for policy simulations

Key Formula / Estimator

ARMA(p,q): $y_t = \sum_{j=1}^{p} \phi_j y_{t-j} + \varepsilon_t + \sum_{i=1}^{q} \theta_i \varepsilon_{t-i}$

VAR(p): $\mathbf{y}_t = \mathbf{A}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{u}_t$

Granger: x Granger-causes y if lags of x jointly significant in VAR.

Watch Out For

- ARIMA requires stationarity first (difference if I(1))
- Model identification: ACF/PACF, then AIC/BIC
- VAR: all series must be stationary (or use VECM if cointegrated)
- Granger causality is predictive, not structural causality
- Impulse responses depend on Cholesky ordering (economic theory)
- Over-parameterised VAR: reduce lag length; LASSO-VAR for large systems

Researcher Checklist

- ☐ Confirm stationarity (ADF) before ARMA/VAR
- ☐ Inspect ACF/PACF for p, q guidance
- ☐ Estimate candidate ARIMA models; compare AIC/BIC
- ☐ Check residuals: Ljung–Box test, ACF of residuals
- ☐ VAR: select lag by AIC/BIC/HQ criteria
- ☐ Test residuals for autocorrelation (Portmanteau)
- ☐ Granger causality: Wald test on lag block
- ☐ Report impulse responses with confidence bands

16. Simultaneous Equations Models

When to Use

- Endogenous variables determined jointly (supply & demand)
- Each equation has a causal interpretation
- IV/2SLS is insufficient (multiple endogenous vars)
- Macro structural models; wage–employment systems

Watch Out For

- Under-identified equation: cannot be estimated
- OLS on structural equation is inconsistent
- Just-identified: 2SLS = IV; over-identified: test with J-test
- 3SLS more efficient but inconsistency spreads across equations
- Normalisation rule: assign one variable as LHS per equation

Key Formula / Estimator

Order condition (necessary):
excluded exogenous $\geq$ # endogenous regressors.

Rank condition (sufficient): rank of restricted coefficient matrix $= G - 1$.

2SLS equation by equation; **3SLS** for system efficiency.

Researcher Checklist

- ☐ Write structural form explicitly (each equation)
- ☐ Check order condition for each equation
- ☐ Verify rank condition before estimation
- ☐ List all exogenous/excluded variables
- ☐ Estimate each equation by 2SLS
- ☐ Sargan test for over-identified equations
- ☐ Compare 2SLS vs. OLS to gauge endogeneity bias
- ☐ Consider 3SLS if system efficiency matters

17. Diagnostics at a Glance — Master Reference Table

Problem	Test	H_0	Stat	Remedy
Heteroscedasticity	Breusch–Pagan	Homoscedastic	χ^2	HC1 robust SEs
Heteroscedasticity	White	Homoscedastic	χ^2	HC1 robust SEs
Serial correlation	Breusch–Godfrey	No AR(p)	F / χ^2	HAC / C–O
Serial correlation	Durbin–Watson	No AR(1)	DW	Add lags / HAC
Multicollinearity	VIF	—	> 10	Drop / PCA / Ridge
Multicollinearity	Condition κ	—	> 30	Same as above
Misspecification	RESET	Correct form	F	Add nonlinear terms
Unit root	ADF	Unit root	t	Difference / ECM
No cointegration	Engle–Granger	Not cointegrated	t	First differences
Endogeneity	Wu–Hausman	Exogenous	χ^2 / F	2SLS
Weak instrument	1st-stage F	Strong instrument	$F < 10$	Find better IV
Overidentification	Sargan/Hansen	Valid instruments	χ^2	Check instruments
Panel: RE vs FE	Hausman	RE consistent	χ^2	FE if $p < 0.05$
Structural break	Chow	No break	F	Split sample / dummies
Normality	Jarque–Bera	Normal residuals	χ^2_2	Check outliers

18. Model Assumptions Summary

Assumption	OLS (cross-section)	Time Series OLS	Violated by
MLR.1	Linear in parameters	TS.1 Same	Misspecification
MLR.2	Random sample	TS.2 Weak dependence	Non-random, $I(1)$
MLR.3	No perfect collinearity	TS.3 Same	Dummy trap
MLR.4	Zero conditional mean	TS.4 Strict exogeneity	Endogeneity, OVB
MLR.5	Homoscedasticity	TS.5 Same	Heteroscedasticity
MLR.6	Normality of u	TS.6 Same	Outliers, small n
(Panel) FE.1	Strict exogeneity given a_i	—	Lagged dep. var. in FE
(IV) IV.1	Instrument relevance: $\pi_1 \neq 0$	—	Weak instruments
(IV) IV.2	Instrument exogeneity: $\text{Cov}(z, u) = 0$	—	Invalid instrument

19. The Researcher's Pre-Submission Checklist

Before You Submit: Run Through All of These

Data & Design
- ☐ Document data source and time period
- ☐ Check for missing values and their pattern
- ☐ Report sample size and key descriptive stats
- ☐ State identification strategy clearly
- ☐ Confirm exogeneity of key regressors

OLS Baseline
- ☐ Run baseline OLS with full controls
- ☐ Run RESET test (functional form)
- ☐ Report robust (HC1) SEs alongside OLS SEs
- ☐ Check R^2, adjusted R^2, AIC/BIC
- ☐ Plot residuals vs. fitted values

Diagnostic Tests
- ☐ Breusch–Pagan and White (heteroscedasticity)
- ☐ Breusch–Godfrey if time series / panel
- ☐ VIF for each regressor
- ☐ ADF if time series data
- ☐ Hausman if panel data

Robustness & Reporting
- ☐ Run model on subsamples / alternative specs
- ☐ Report placebo / falsification tests
- ☐ Show main table with multiple specifications
- ☐ Discuss direction and magnitude of violations
- ☐ State which assumptions cannot be tested

20. Quick Python / statsmodels Reference

Task	Python snippet
OLS	`smf.ols('y ~ x1 + x2', data=df).fit()`
Robust SEs	`res.get_robustcov_results(cov_type='HC1')`
HAC SEs	`res.get_robustcov_results(cov_type='HAC', maxlags=4)`
Breusch–Pagan	`het_breuschpagan(res.resid, res.model.exog)`
White test	`het_white(res.resid, res.model.exog)`
Breusch–Godfrey	`acorr_breusch_godfrey(res, nlags=4)`
ADF test	`adfuller(series, autolag='AIC')`
VIF	`1 / (1 - smf.ols(f'{x} ~ ...').fit().rsquared)`
2SLS	`IV2SLS.from_formula('y ~ 1 + [x ~ z]', data=df).fit()`
FE panel	`PanelOLS.from_formula('y ~ x + EntityEffects', data=pdata).fit()`
Hausman	`compare(fe_res, re_res)` from `linearmodels`
Logit AME	`logit_res.get_margeff().summary()`
ARIMA	`ARIMA(y, order=(p,d,q)).fit()`
Granger	`grangercausalitytests(data[['y','x']], maxlag=4)`